The ESO Astroquery module: a Jupyter Notebooks walkthrough

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eas 2026 Lausanne

Hands-on session

Please download the slides from below

https://github.com/eso/astroquery_examples/blob/main/assets/eas2026-ucc-eso-astroquery.pdf

https://tinyurl.com/eas2026-eso

What is Astroquery?

Astroquery enables end-to-end analyses

"Astroquery is a set of tools for querying astronomical web forms and databases."

Documentation: astroquery.readthedocs.io

Code and issue tracker: https://github.com/astropy/astroquery

Paper: *Ginsburg, Sipőcz, Brasseur et al 2019.*

Available Services

The following modules have been completed using a common API:

- ALMA Queries (**astroquery.alma**)
- Atomic Line List (**astroquery.atomic**)
- Besancon Queries (**astroquery.besancon**)
- CADC (**astroquery.cadc**)
- CASDA Queries (**astroquery.casda**)
- Cologne Database for Molecular Spectroscopy (CDMS) Queries (**astroquery.linelists.cdms**)
- ESA EUCLID Archive (**astroquery.esa.euclid**)
- ESA Herschel Science Archive (**astroquery.esa.hsa**)
- ESA HST Archive (**astroquery.esa.hubble**)
- ESA Integral Science Legacy Archive (ISLA) (**astroquery.esa.integral**)
- ESA ISO Archive (**astroquery.esa.iso**)
- ESA JWST Archive (**astroquery.esa.jwst**)
- ESA XMM-Newton Archive (**astroquery.esa.xmm_newton**)
- ESASky Queries (**astroquery.esasky**)
- ESO Queries (**astroquery.eso**)
- FIRST Queries (**astroquery.image_cutouts.first**)
- Gaia TAP+ (**astroquery.gaia**)
- GAMA Queries (**astroquery.gama**)
- Gemini Queries (**astroquery.gemini**)
- HEASARC Queries (**astroquery.heasarc**)
- HiPS2fits Service (**astroquery.hips2fits**)
- HITRAN Queries (**astroquery.hitran**)
- IRSA Moving Object Search Tool (astroquery.ipac.irsa.most)
- IRSA Dust Extinction Service Queries (**astroquery.ipac.irsa.irsa_dust**)
- IRSA Image Server program interface (IBE) Queries (**astroquery.ipac.irsa.ibe**)
- IRSA Queries (**astroquery.ipac.irsa**)
- JPL Spectroscopy Queries (astroquery.jplspec)
- MAGPIS Queries (**astroquery.magpis**)
- MAST Queries (**astroquery.mast**)
- CDS MOC Service (**astroquery.mocserver**)
- Minor Planet Center Queries (**astroquery.mpc**/astroquery.solarsystem.MPC)
- NASA ADS Queries (**astroquery.nasa_ads**)
- NED Queries (**astroquery.ipac.ned**)
- NIST Queries (**astroquery.nist**)
- NVAS Queries (**astroquery.nvas**)
- SIMBAD Queries (**astroquery.simbad**)
- Skyview Queries (**astroquery.skyview**)
- Splatalogue Queries (**astroquery.splatalogue**)
- SVO Filter Profile Service Queries (**astroquery.svo_fps**)
- UKIDSS Queries (**astroquery.ukidss**)
- Vamdc Queries (**astroquery.vamdc**)
- VizieR Queries (**astroquery.vizier**)
- VO Simple Cone Search (**astroquery.vo_conesearch**)
- VSA Queries (**astroquery.vsa**)
- xMatch Queries (**astroquery.xmatch**)

The ESO module delivers complex datasets quickly

“Simple python functions to script your access to the ESO archive via TAP.”

ESO astroquery fork: https://github.com/eso/astroquery

ESO example notebooks: https://github.com/eso/astroquery_examples

- *The ESO astroquery module allows users to search for raw and reduced data, retrieve metadata, access catalogues, and download data products. Proprietary access is also supported.*

WDB (“Old”)

- Limited query interfaces with predefined queries
- Users can only enter constraints on predefined queries
- No direct query language
- Users cannot modify underlying queries
- Catalogues not accessible
- Simplified access for specific, common queries

TAP (“New”)

- Full read-only access to ESO databases
- Users can write their own custom queries
- Uses ADQL 2.0 (SQL-based, VO standard)
- Users fully control query structure
- Catalogues accessible via dedicated interface
- Advanced, flexible querying for complex data retrieval

- *The **new** ESO module enables faster, larger and more complex dataset analyses by using the the Table Access Protocol (**TAP**), an International Virtual Observatory Alliance (**IVOA**) standard.*

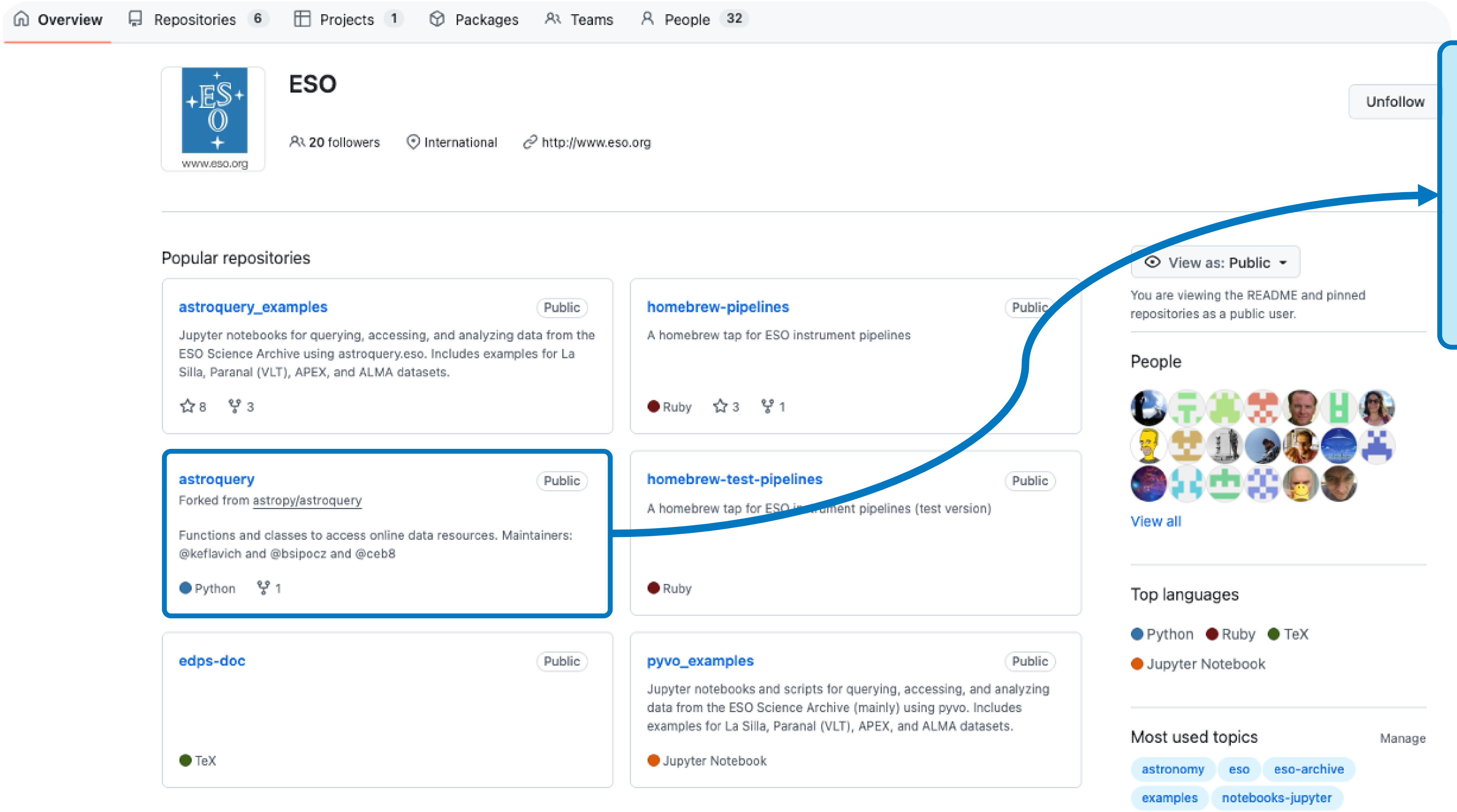
Functionality

- List instruments and data releases.
- Instrument-specific and instrument-independent raw data search.
- Search across / by data releases.
- Automatic calibration selection (raw).
- Get extended FITS header information.
- Download datasets by their identifiers.
- Authenticated access to your assets.
- Run custom / complex ADQL queries.
- Retrieve atmospheric conditions data.

→ Automate all the above ←

Where to get the new ESO astroquery module and examples?

Get the ESO module and examples from GitHub



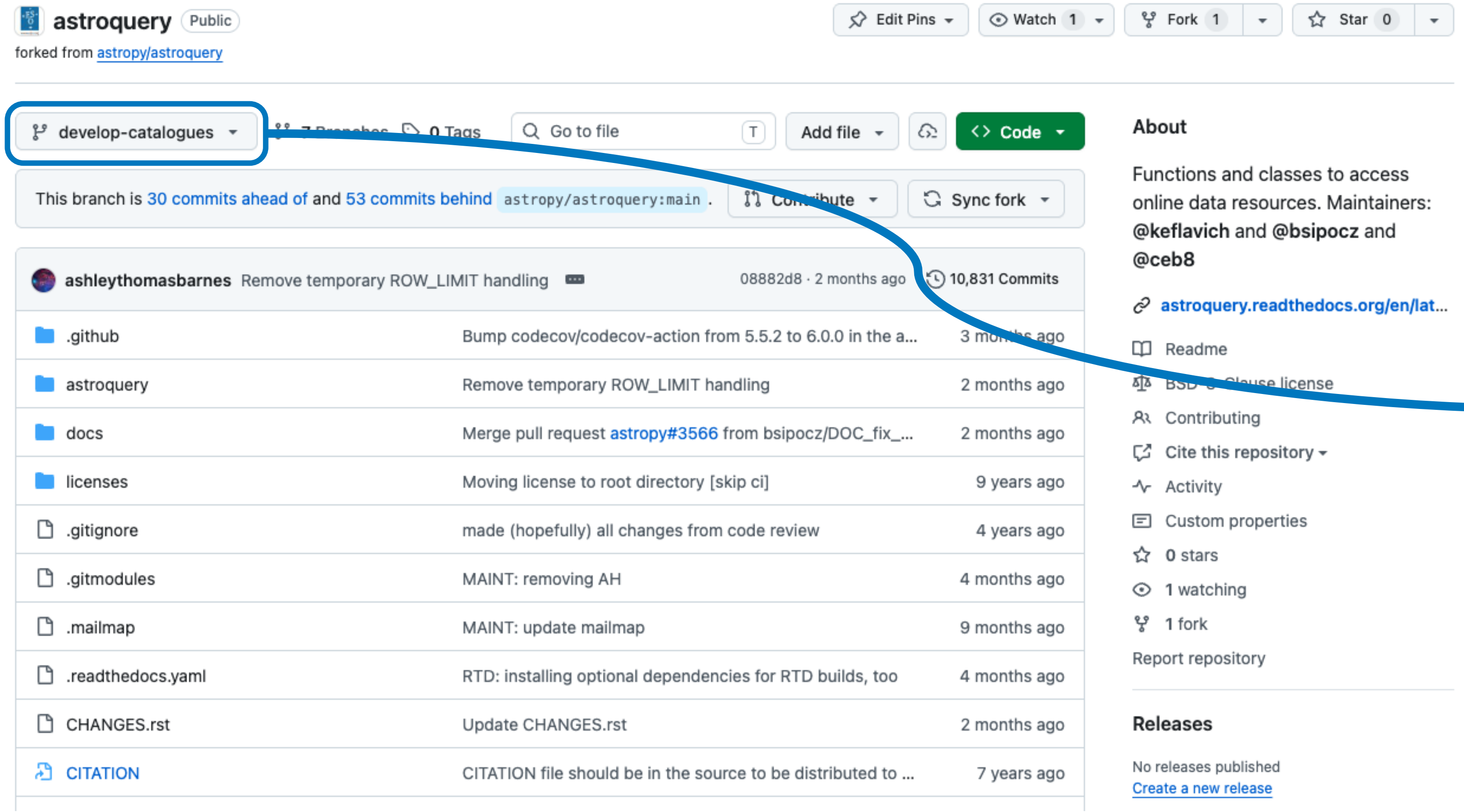
astroquery

- Fork of the official astroquery repository
- Latest ESO features not yet part of an official astroquery release

github.com/eso

 https://tinyurl.com/eas2026-eso

Get the ESO module and examples from GitHub



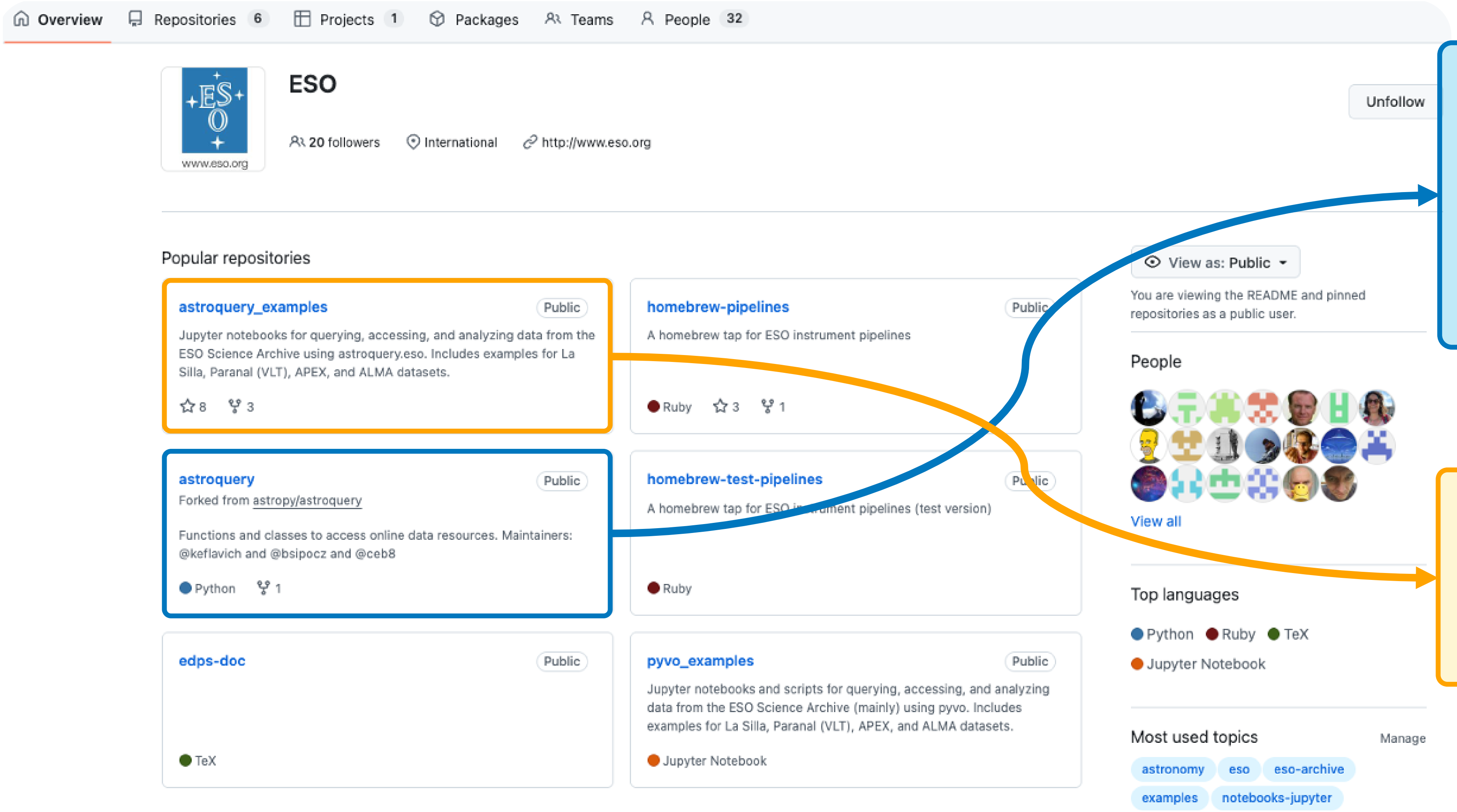
astroquery

- **Important** for this example session we need the catalogue development version of astroquery

pip install "git+https://github.com/eso/astroquery.git@develop-catalogues"

github.com/eso

Get the ESO module and examples from GitHub



astroquery

- Fork of the official astroquery repository
- Latest ESO features not yet part of an official astroquery release

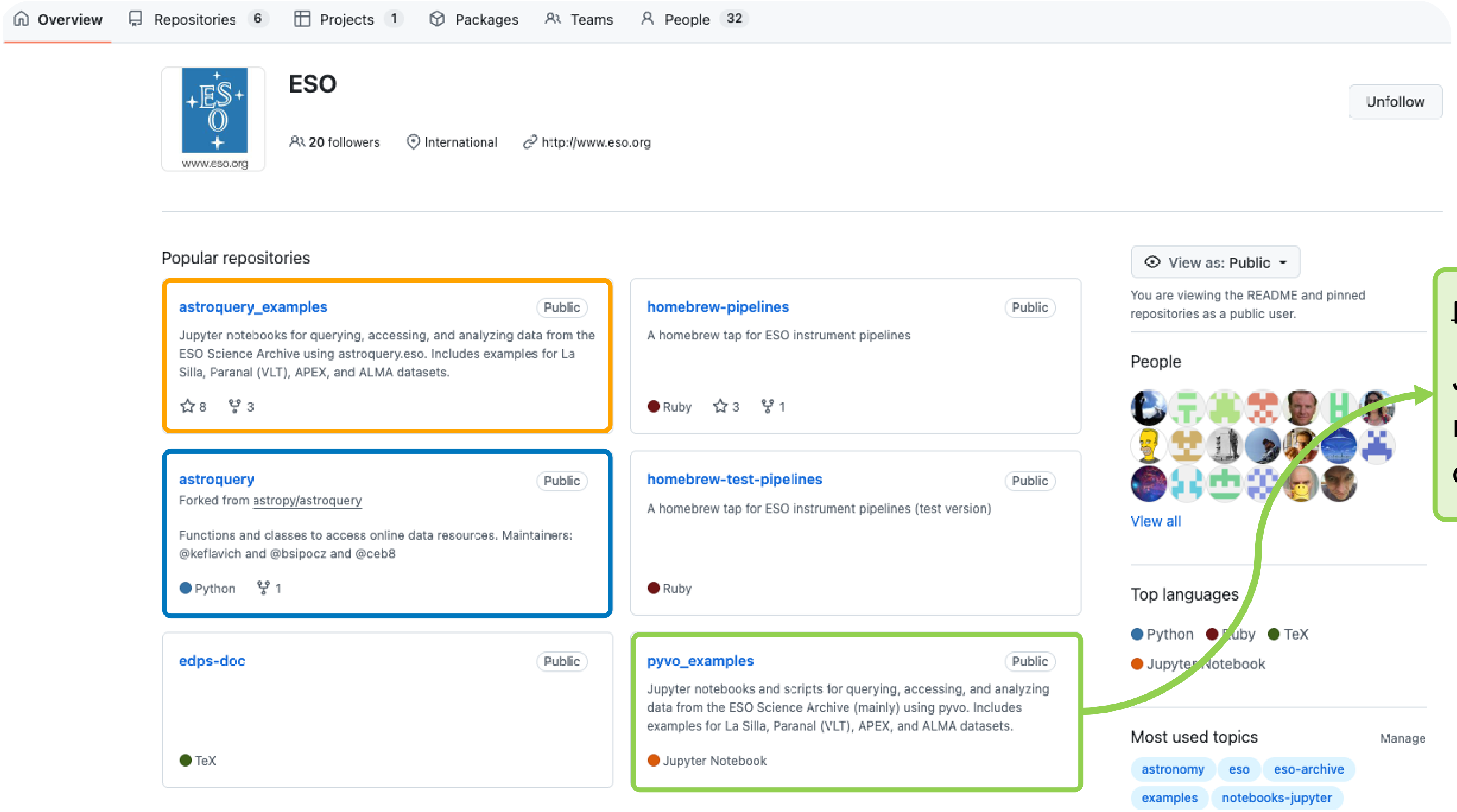
astroquery_examples

- Jupyter notebooks with tutorials and real application examples

https://tinyurl.com/eas2026-eso

github.com/eso

Get the ESO module and examples from GitHub



pyvo_examples

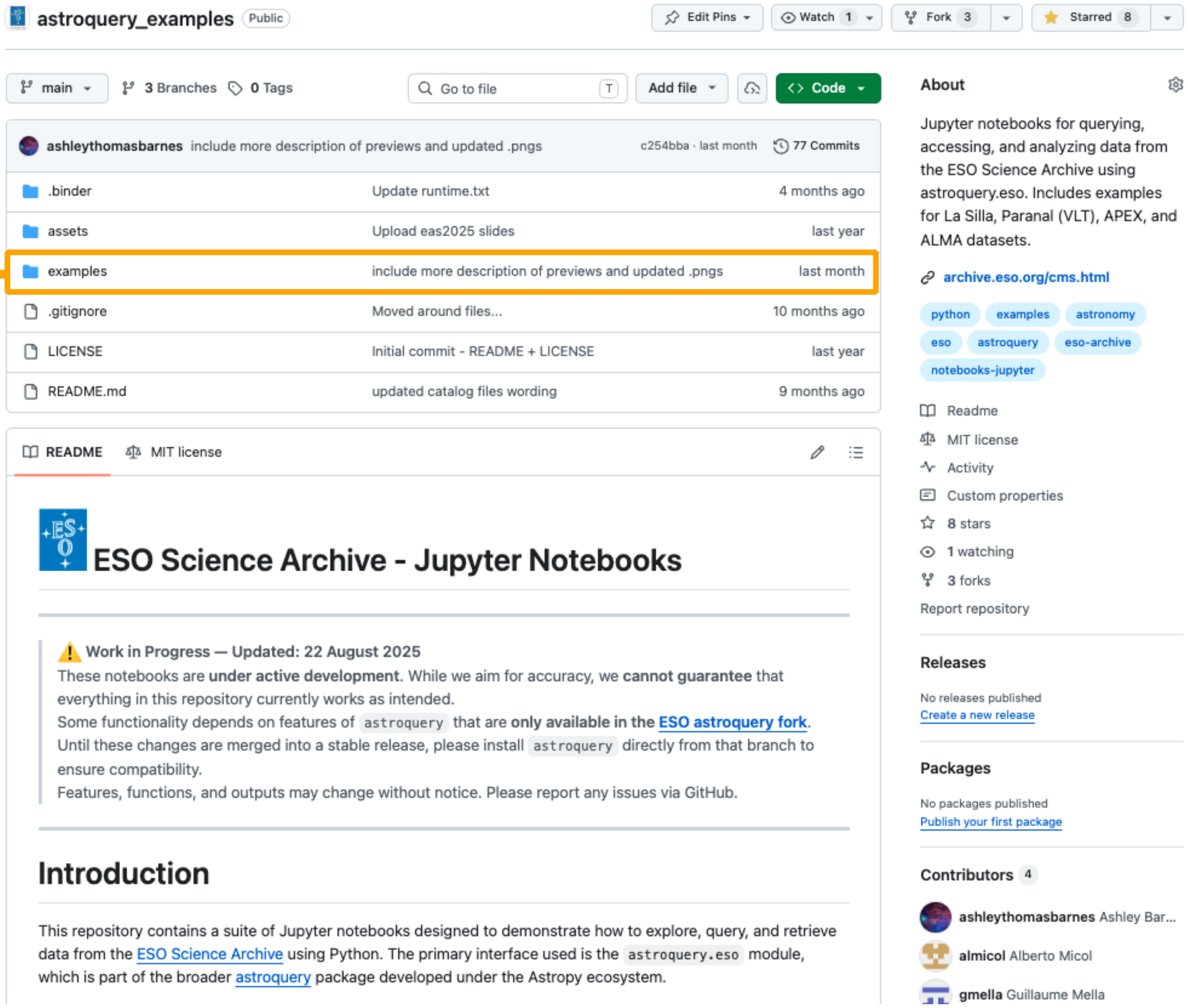
Jupyter notebooks with tutorials and real application examples making use of pyvo module.

github.com/eso

Get the ESO module and examples from GitHub

astroquery_examples

- Jupyter notebooks with tutorials and real application examples



github.com/eso

Get the ESO module and examples from GitHub

Name	Last commit message	Last commit date
..		
advanced	update dawn analysis with comments	9 months ago
case_studies	include more description of previews and updated .pngs	last month
simple	updated catalog files wording	9 months ago
README.md	Updated to highlights -> case_studies	10 months ago

Introduction to basic functionality

- Query the archive for raw* and reduced data
- Perform a simple cone search
- Obtain extended information on data products
- Download datasets from the archive

*Either instrument-specific or generic

Tutorials on basic functionality

- APEX products - reduced, raw, quick looks
- Searching via ADQL (SQL-like syntax)
- Downloading datasets by identifier
- Searching a source by name

Real life examples and use cases

- Find the closest observation to a given time
- Investigate ESO data using Aladin
- Download raw data for a reduced data product
- Examine Spectral Data as function of time

github.com/eso

Get the ESO module and examples from GitHub

main · astroquery_examples / examples / simple /

Go to file · Add file

ashleythomasbarnes Add instrument-specific query example · 9351e0d · 8 minutes ago · History

Name	Last commit message	Last commit date
..		
00_quickstart.ipynb	Add instrument-specific query example	8 minutes ago
01_authentication.ipynb	checks for simple examples	last month
02_observations_query_reduced_by_position.ipynb	checks for simple examples	last month
03_observations_query_raw_by_position.ipynb	checks for simple examples	last month
04_observations_query_by_program_id.ipynb	checks for simple examples	last month
05_observations_query_observation_apex.ipynb	checks for simple examples	last month
06_observations_query_tap.ipynb	Add TAP cone, polygon and spatial examples	last month
07_observations_download_data.ipynb	checks for simple examples	last month
08_catalogs_query_with_constraints.ipynb	checks for simple examples	last month
09_catalogs_query_by_position.ipynb	checks for simple examples	last month
10_catalogs_query_tap.ipynb	updates to query_tap	last month
README.md	Replace introduction notebook with quickstart	16 minutes ago

Take a test drive
– Hands-on session –

Take a test drive on the cloud

README MIT license

Introduction

This repository contains a suite of Jupyter notebooks designed to demonstrate how to explore, query, and retrieve data from the ESO Science Archive using Python. The primary interface used is the `astroquery.eso` module, which is part of the broader astroquery package developed under the Astropy ecosystem.

These notebooks walk through both introductory and advanced examples that are relevant for a wide range of scientific use cases — from downloading raw and reduced data products, to analyzing spectra and imaging datasets retrieved from ESO facilities at La Silla and Paranal (including the Very Large Telescope - VLT - and the Very Large Telescope Interferometer - VLTI), as well as from submillimeter and radio observatories -- APEX and ALMA.

Whether you're new to archival astronomy or already familiar with ESO's observing systems, these examples provide practical guidance for integrating archive queries into reproducible, scriptable workflows.

📚 Examples

All notebooks live under `examples/` and are grouped by purpose:

```
examples/
├── simple/          # 🌱 Minimal, single-feature recipes (copy/paste friendly)
├── advanced/        # 🚀 Scenario-driven workflows combining multiple functions
└── case_studies/    # 🌌 Curated case studies for specific Phase 3/data collections
```

See detailed indexes inside each folder:

- `examples/simple/README.md`
- `examples/advanced/README.md`
- `examples/case_studies/README.md`

▶️ Quick Start: launch binder

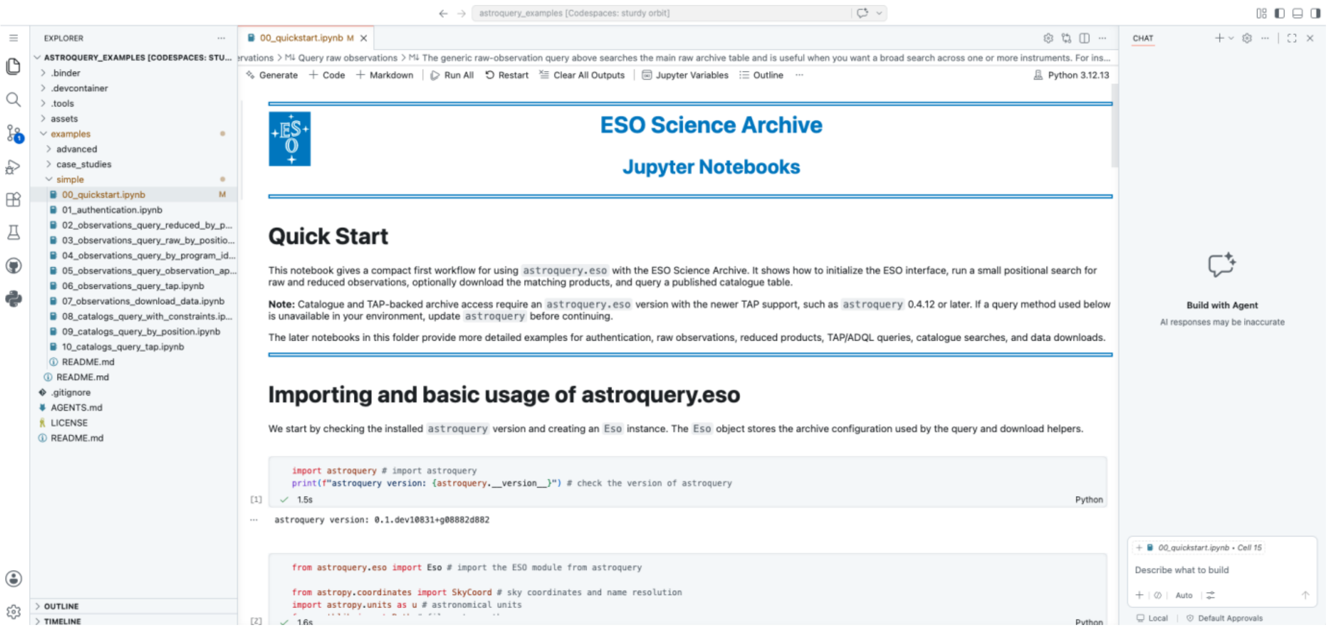
To get started, you can load and initialize the ESO archive interface directly in your notebook:

```
from astroquery.eso import Eso  # Import the ESO module from astroquery
eso = Eso()                     # Create an instance of the ESO class
eso                             # Display the class instance (e.g. check login status, default
```

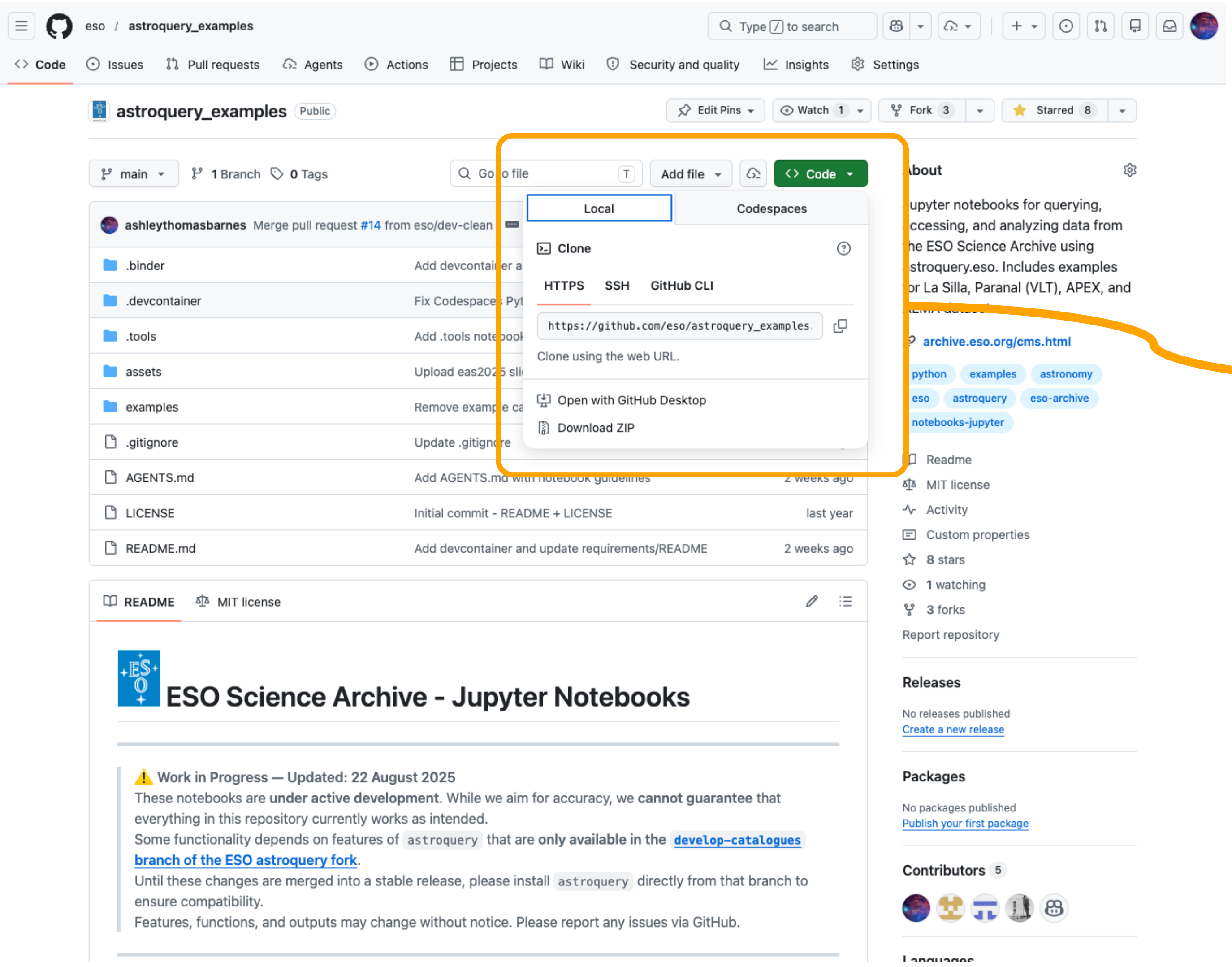
A quick-start notebook can be found in `examples/simple/00_quickstart.ipynb`.

binder

Scroll down the github page and click here – **may take a little time to load up, so be patient…**

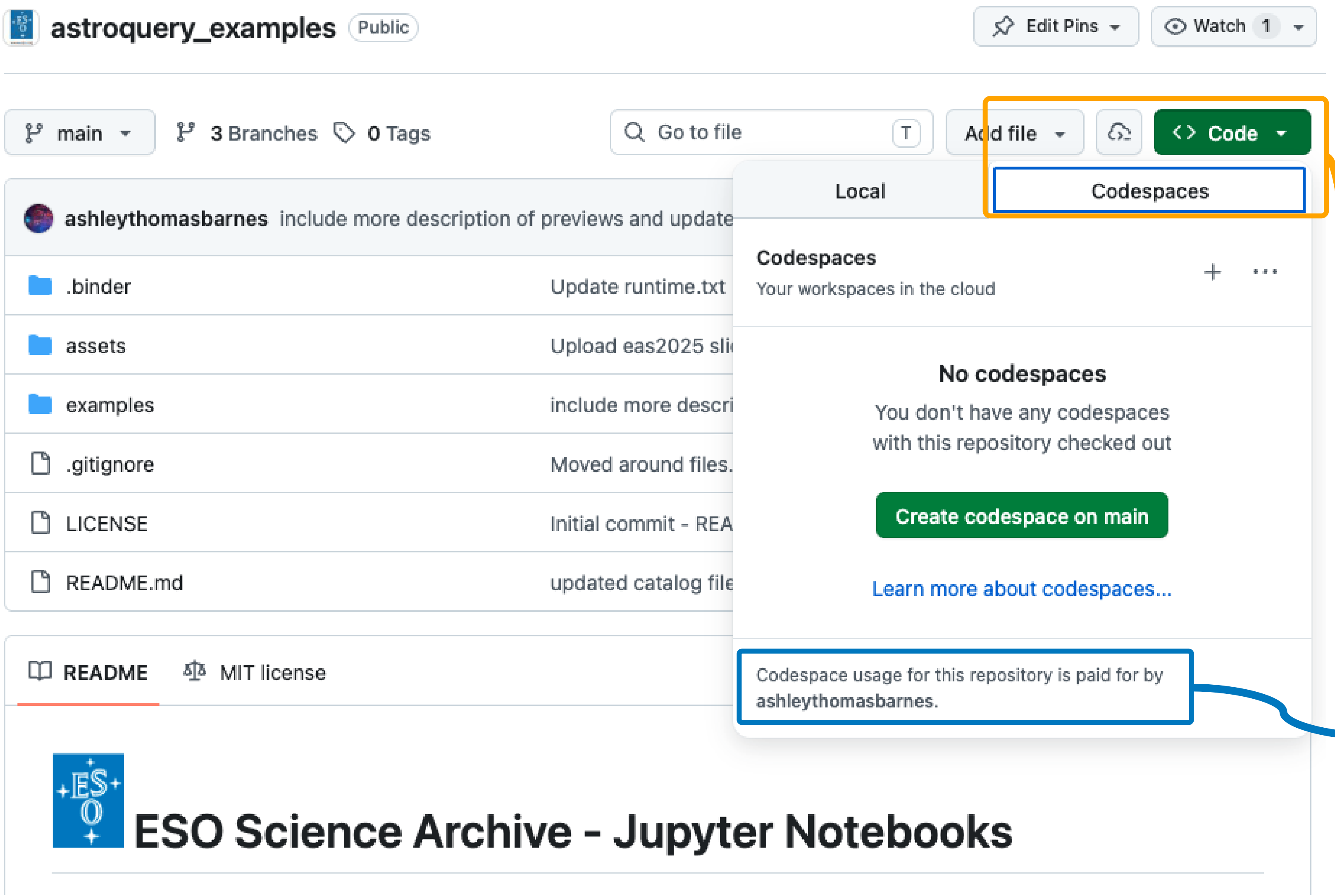


Take a test drive on the cloud



You can also download locally and run the notebooks on your machine – **but make sure you have the correct version of astroquery installed**

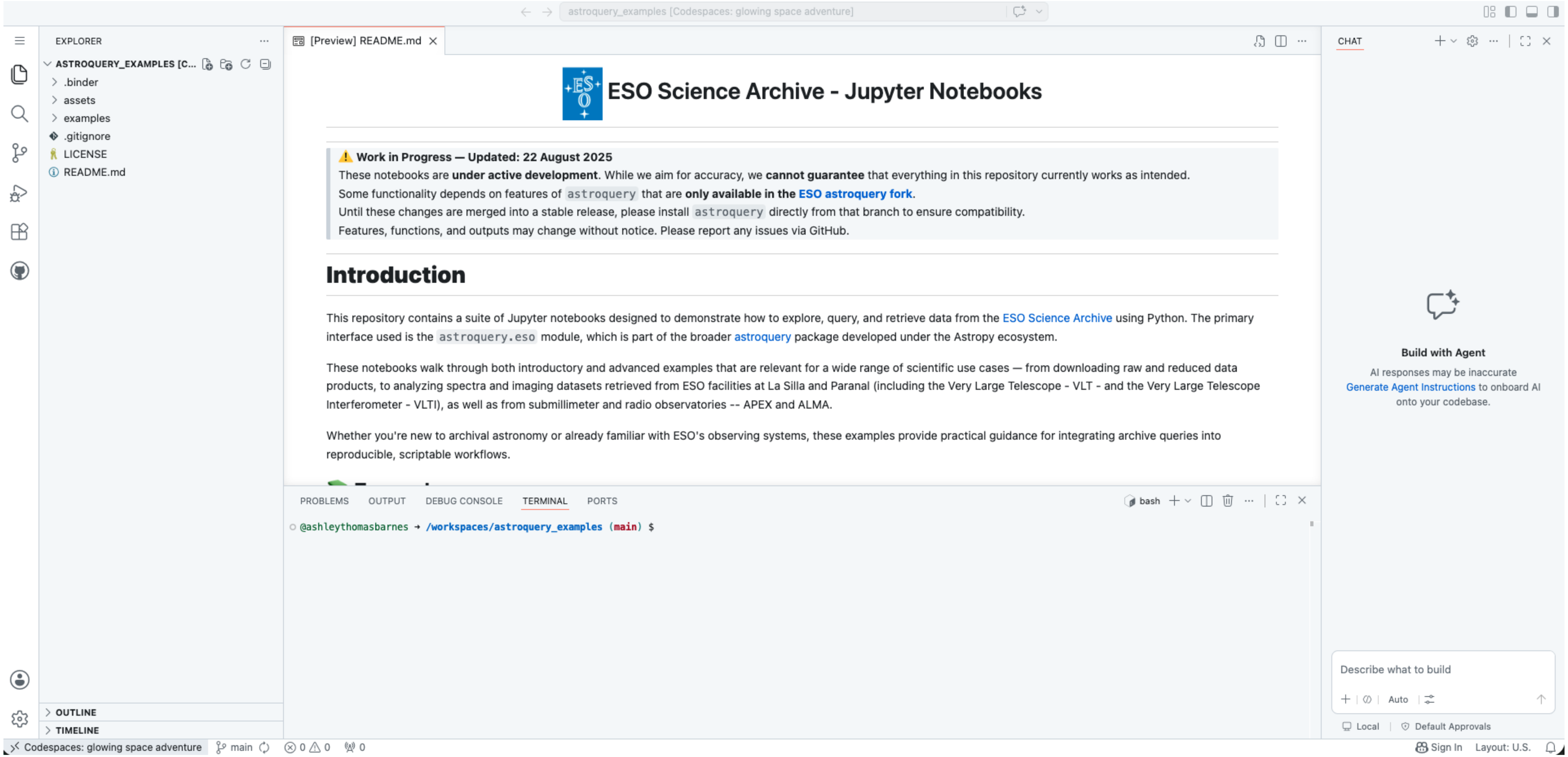
Take a test drive on the cloud



For this example we will use github codespaces – which you can find by clicking on “code” and then “create codespace on main” - be patient this can take a little time to load…

Don’t worry about this – its free if you have a github account!

Take a test drive on the cloud



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Thank you!

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